

```

import numpy as np
import pandas as pd
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler
import matplotlib.pyplot as plt

def main():
    data = {
        'Feature1': [2.5, 0.5, 2.2, 1.9, 3.1, 2.3, 2, 1, 1.5, 1.1],
        'Feature2': [2.4, 0.7, 2.9, 2.2, 3, 2.7, 1.6, 1.1, 1.6, 0.9]
    }

    df = pd.DataFrame(data)

    X = df.values

    sc = StandardScaler()
    X_scaled = sc.fit_transform(X)

    pca = PCA(n_components=1)
    X_pca = pca.fit_transform(X_scaled)

    print("Original Data Shape:", X.shape)
    print("Reduced Data Shape:", X_pca.shape)
    print("PCA Result:\n", X_pca)

    plt.scatter(X_pca, np.zeros_like(X_pca))
    plt.title("PCA - 1D Projection")
    plt.xlabel("Principal Component 1")
    plt.show()

if __name__ == "__main__":
    main()

```

Original Data Shape: (10, 2)

Reduced Data Shape: (10, 1)

PCA Result:

```

[[ 1.08643242]
 [-2.3089372 ]
 [ 1.24191895]
 [ 0.34078247]
 [ 2.18429003]
 [ 1.16073946]
 [-0.09260467]
 [-1.48210777]
 [-0.56722643]
 [-1.56328726]]

```

